Data Mining Techniques.
Assignment 1
Group Project (2 People)



In this assignment you will work with data from a well-known home rental application. Specifically, you are given the data for the Athens area for 3 months of 2019 and of 2023. The data is in csv format and you will use Python to answer the questions below.

Part A

Question 1: Data Analysis (Data exploration)

The data you are given is organized in folders (e.g. April, March, February). Each folder contains different csv files that you must combine and merge appropriately using python and pandas. Specifically, you will need to create two different csv files (one for 2019 and one for 2023) that will contain the following columns.

id
zipcode
transit
Bedrooms
Beds
Review_scores_rating
Number_of_reviews
Neighbourhood
Name
Latitude
Longitude
Last_review
Instant_bookable
Host_since

Host_response_rate
Host_identity_verified
Host_has_profile_pic
First_review
Description
City
cancellation_policy
Bed_type
Bathrooms
Accommodates
Amenities
Room_type
Property_type
price
Availability_365
Minimum_nights

In each of these two files (name them train_2019.csv and train_2023.csv) you will need to examine whether there are missing data. Decide how you will handle them and either remove them or fill them in appropriately in the corresponding train files. In addition, you will need to examine whether there are also outliers (for example if the number of rooms is 7000!) and handle these cases accordingly as well.

Next, you are asked to answer the questions below using plots, histograms, heat maps, etc. with either matplotlib or seaborn or any other library you wish. The analysis should first be done for each year separately, except for **1.9**, which can be done for only one year of your choice.

1.1 What is the most frequent room_type in your data?
1.2 Create a plot or plots that show the trend of prices over the 3-month period.
1.3 Which are the top 5 neighbourhoods with the most reviews?
1.4 Which neighbourhood has the most property listings?
1.5 How many listings are there per neighbourhood and per month?
1.6 Plot the histogram of the neighborhood variable.
1.7 What is the most frequent room type (room_type) in each neighbourhood (neighborhood)?
1.8 Which is the most expensive room type?
1.9 Use the Folium Map library with the latitude/longitude columns and display on a map, for a month of your choice, the properties, and in the popups on the map choose any other information you want to be shown for the property (e.g. bed_type, room_type, transit etc).
1.10 Create different wordclouds with the data from the neighbourhood, transit, description, last_review columns.

1.11 In this question we will make use of the **amenities** column. If you print the unique values this column contains, you will see that it would be better to simplify these values. For example, we could group them into the following categories: *kitchen, accesibility, Electricity_and_Technology, facilities, kids_friendly, security, services*. For example, all the values Breakfast, Cooking basics, BBQ, grill, oven, Coffee maker, etc. can be replaced by the word *kitchen*. Using the same logic you can think about how to replace the remaining values in the amenities column. The result will be that this column has fewer and distinct values that could in turn be used for extracting information (e.g. in question 1.13). The categories we suggest are indicative; you can use fewer categories (or more, or even different ones, as long as the goal of simplification is not lost). Then, after you have simplified this column, create a histogram showing the distribution of the new values.
1.12 First compute the average price per neighbourhood (the price column will be used, and so that we compare like with like, keep only the rooms that accommodate two people). Show in a plot the neighbourhoods sorted by their average price. Then classify the neighbourhoods into 3 groups (very expensive, moderate, affordable) based on the average price of the rooms.
1.13 What other information could you extract from the data you are given? Think of **3 additional** questions about the Athens area and display the results (you may combine as many columns as you like from any year).
1.14 Recent laws on Airbnb single out individuals who list three or more properties for rent (or sublet) on a digital platform. Create a table with the information (host_id, num_host_listings) containing the 10 hosts with the most properties.
1.15 Given that you studied the above for two different years, what could your own conclusion be about the Athens area regarding short-term rentals? For example, are there more / different neighbourhoods between the two years? Did prices increase? Did the expensive neighbourhoods remain expensive? In other words, try to make a comparison between the two years based on the plots you have presented in the questions above.

Question 2: Recommendation System

In this question you will need the columns: 'id', 'name', 'description'.

The goal is to extract useful information from this data and try to build a program that produces suggestions (recommendations) for the Athens area. In the first question you have already created the wordclouds for the description column. In this question, remove the stop words, experiment with the wordcloud parameters and identify the most characteristic words that the visitor uses for the Athens area. Next, create a new column that will contain the concatenation of the name and description columns (fill NA with NULL). Answer the following:

1. Create the TF-IDF (Term Frequency - Inverse Document Frequency) matrix of the unigrams and bigrams for the new column (use the stop_word parameter of the TfidfVectorizer).
2. Cosine Similarity: This metric computes the similarity between two vectors x,y, using the angle between them (when the angle is 0 it means that x and y are equal, apart from their length). Iterate over the TF-IDF matrix and compute the similarity of each property with the rest. Store the 100 most similar properties in a python dictionary.
3. Prediction: Create a function that takes as input an id and an integer N, and returns the N most similar properties.

   ```
   recommend(item_id = 4085439, num = 5)
   ```

   The output of the function should have the following format:

   Recommending 5 listings similar to Studio

   Recommended: NAME
   Description: DESCRIPTION
   (score:0.12235188993161432)

   Recommended: NAME
   Description: DESCRIPTION
   (score:0.12235188993161432)
   ……..
4. Words that frequently appear together with other words (collocation). Use the BigramCollocationFinder to find 10 words that “tend” to appear together frequently.

Deliverable:

The assignment can be carried out **individually or in groups of 2 people (preferably work in a group!)** .

You will upload to eclass a folder of the form sdixxxx.zip (where sdi is the student ID of one of the group members) which will contain only your code in **Ipython notebook** format (note: you do not need to upload the train.csv files as well). You may also have more than one notebook if it suits you.

The notebook must have been “run” so that the results of your work are visible. The notebook also serves as the complete report for your assignment (you will not submit anything in doc, pdf), design it carefully, and remember to write which question is answered in each cell. Clarifications about the assignment will be given via e-class.